

MS803 Nano Oil 1

Chemical Safety Data Sheet (SDS)

Document No.: ZG-SDS20170321-MS803-AC-1500

Section 1 Chemical Product and Company Identification

Chemical product

Product name: MS803 Nano-Oil 1 (AC-1500)**Recommended use:** For cutting and grinding of ferrous metals.

Company identification

Region	Supplier (company name / address / phone)	Emergency phone
China, Shanghai	Supplier name: Shanghai Binxin Optoelectronics Technology Co., Ltd.	Supplier address: Building 10, Lane 16299, Puwei Hig

Classification: This product is not classified as a hazardous substance.

Section 2 Hazards Identification

Hazard label: —**Health hazards:** Excessive exposure may cause irritation to the eyes, skin, or respiratory system.**Environmental hazards:** No significant hazard.**Fire and explosion hazards:** Non-flammable liquid; no significant explosion hazard.**Composition:** Mixture

Section 3 Composition / Information on Ingredients

Name	CAS Reg. No.	Content, %
Refined mineral oil	—	75–90
Sulfurized fat	—	5–10
Synthetic ester	—	5–15
Rust inhibitor	—	0.05–1

Section 4 First Aid Measures

Inhalation: Use suitable respiratory protection. If respiratory irritation, dizziness, nausea, or unconsciousness occurs, seek medical attention immediately. If breathing stops, use mechanical ventilation equipment, or administer mouth-to-mouth artificial respiration.**Skin contact:** Wash the affected area with soap and water. Remove contaminated clothing. Contaminated clothing should be washed before reuse.**Eye contact:** Rinse thoroughly with water. If irritation occurs, seek medical aid.**Ingestion:** Seek medical treatment immediately. Do not induce vomiting. Remain at rest and seek medical care promptly.

Section 5 Firefighting Measures

Extinguishing media

Suitable extinguishing media: Use water fog, foam, dry chemical (dry powder), or carbon dioxide (CO₂).**Unsuitable extinguishing media:** Direct water stream.

Firefighting

Firefighting instructions: Evacuate the area. Prevent controlled fire or diluted runoff from flowing into rivers, sewers, or drinking water sources.

Firefighters should use standard protective equipment; in confined spaces, use self-contained breathing apparatus. Use water spray to cool surfaces exposed to fire and to protect personnel.

Fire hazards: Firefighters should consider the protective equipment described in Section 8.**Hazardous combustion products:** Incomplete combustion products, carbon dioxide.**Hazard characteristics:** No significant hazardous characteristics.

Section 6 Accidental Release Measures

Ground spill: Do not touch or walk through the spilled product. Avoid flow into waterways, sewers, basements, or confined areas. Cover with dry soil, sand, or other non-combustible material and transfer to containers. Recover by pumping or using suitable absorbents.

Water spill: Use oil containment booms to limit the spill extent. Notify the port or relevant authorities. Skim from the surface or remove with suitable absorbents. If permitted by local agencies and environmental departments, allow contaminants to settle and/or appropriately use dispersants in open waters. Consult relevant experts regarding the specific disposal requirements for all recovered material to ensure compliance with local waste disposal regulations.

Section 7 Handling and Storage

Handling precautions: This product can accumulate static charge, which may produce electric sparks; sparks can ignite any flammable vapors from the liquid or residues. Appropriate bonding or grounding measures must be taken. Consult applicable local standards as a guide.

Loading temperature: Ambient

Transport temperature: Ambient

Storage precautions: Keep containers tightly closed. Handle containers with care. Store in a cool, well-ventilated place. Storage containers should be grounded. Fixed storage containers must be bonded or grounded. Do not pressurize, cut, heat, or weld containers. Empty containers may contain product residue; do not reuse empty containers before professional cleaning or reconditioning.

Storage temperature: Ambient

Storage pressure: Atmospheric

Compatible materials and coatings: Carbon steel; stainless steel; polyethylene; polypropylene; polyester.

Incompatible materials and coatings: Natural rubber; butyl rubber; polystyrene; EPDM rubber.

Section 8 Exposure Controls / Personal Protection

Engineering controls: Provide adequate local exhaust ventilation in tightly enclosed areas. Provide emergency shower and eyewash equipment on site.

Personal protection

Respiratory protection: If airborne concentrations exceed the limits, wear a filter-type gas mask (half mask).

Hand protection: Use chemical-resistant gloves (nitrile gloves).

Eye protection: If contact is possible, it is recommended to use protective goggles with side shields.

Skin and body protection: Use chemical/oil-resistant antistatic work clothing.

Other protection: Maintain good personal hygiene, e.g., wash hands after handling this product and before eating, drinking and/or smoking. Periodically clean work clothing and protective equipment to remove contaminants. Discard contaminated clothing and shoes that cannot be cleaned.

Section 9 Physical and Chemical Properties

Physical state: Liquid

Appearance: Colorless / light-colored transparent

Odor: Characteristic odor

Odor threshold: No data

Relative density (@ 20 °C): 0.81

Flash point: 105 °C

pH value: Not applicable

Melting point: < -9 °C

Boiling point: > 250 °C

Auto-ignition temperature: > 250 °C

Decomposition temperature: No data

Relative vapor density (air = 1): —

Saturated vapor pressure: No data

Log octanol/water partition coefficient: No data

Solubility: Soluble in most organic solvents such as alcohols, ethers, benzene, chloroform, etc.

Other physical properties: None

Section 10 Stability and Reactivity

Stability: The product is stable under normal conditions.

Conditions to avoid: Avoid excessive heat. High-energy ignition sources.

Materials to avoid: Strong oxidizers, acids, alkalis.

Hazardous decomposition products: CO, CO₂, H₂O.

Hazardous polymerization: Hazardous polymerization will not occur.

Section 11 Toxicological Information

	LD ₅₀ (rat)	LD ₅₀ (mouse)	LD ₅₀ (rabbit) / dermal	Other
Acute toxicity	> 5000 mg/kg	> 5000 mg/m ³	> 5000 mg/kg	None
Subacute and chronic toxicity	No data			
Irritation	No data			
Other	No data			

Section 12 Ecological Information

Ecotoxicity: This finished product is considered harmful to aquatic organisms.

Biodegradability: Biodegradable.

Non-biodegradability: Not significantly non-biodegradable.

Bioaccumulation / bioconcentration: No data.

Other adverse effects: No data.

Section 13 Disposal Considerations

Disposal recommendations: This product is not suitable for disposal by deep burial, nor for discharge into public sewers, drainage systems, or natural rivers.

This product can be burned directly in suitable equipment, or incinerated under supervision at very high temperatures to prevent undesirable combustion products. Empty drums should be recycled, regenerated, or disposed of by qualified or licensed agencies. In all cases, exercise caution to ensure compliance with national laws and local regulations.

Disposal precautions: Dispose of in accordance with relevant government regulations.

Section 14 Transport Information

Packaging identification	—	Packing category	II
Packaging method	Refer to national and local regulations before disposal.		
Transport precautions	Handle with care during loading and unloading to prevent damage to packaging and containers.		
Land transport	Not regulated for land transport.		
Sea transport	Not regulated for sea transport.		
Air transport	Not regulated for air transport.		

Section 15 Regulatory Information

Provisions covering the safe use, production, storage, transport, and handling of chemical hazardous substances:

- Regulations on the Safety Management of Chemical Hazardous Substances — effective December 1, 2011
- General Rules for Classification and Hazard Communication of Chemicals, GB 13690-2009 — effective May 1, 2010
- Safety Data Sheet for Chemical Products — Content and Order of Sections, GB 16483-2008 — effective February 1, 2009
- Rules for Preparation of Chemical Safety Labels, GB 15258-2009 — effective May 1, 2010
- Law on the Prevention and Control of Water Pollution / Law on the Prevention and Control of Environmental Pollution by Solid Waste

Section 16 Other Information

Note: This information applies only to the specified substance and may not apply when this substance is mixed with other substances or in the course of use. This information is based on all the knowledge of Zhigao Chemical Materials Co., Ltd. regarding this product as of the date stated and is believed to be accurate and reliable. However, the company makes no warranty or guarantee, express or implied, as to the accuracy, reliability, or completeness of this information. Users themselves must be responsible for the applicability and completeness of this information based on their own applications.

Prepared by: Shanghai Binxin Optoelectronics Technology Co., Ltd.

Prepared by (person): Cai Yuehua

Date prepared: 2019-10-16

Date revised: 2026-03-02

Version: 4th edition